

Procedures

Application time per wristband

- Give the participant the wristband and the same instruction used for everyone
- Let them do one practice run
- Reset the wristband
- Start the timer when they first touch the mechanism
- Watch them apply the wristband
- Stop the timer when the wristband is correctly closed
- Mark whether the trial was successful or not
 - Properly closed
 - Proper fit (two fingers of clearance)
- Repeat for three scored trials per participant
- Repeat for every concept
- Calculate median success time for each concept
- Calculate first-attempt success rate for each concept

Complexity of actions per step

- Let participant do one practice run
- Observe one full scored application trial
- Break down what they did into clear user-visible steps
- Count elemental actions in each step
 - Grasp strap
 - Pull strap
 - Align tab with slot
 - Press snap closed
 - Hold buckle steady
 - Check fit with fingers
- Score each step with this rubric:
 - 1 = single obvious gross-motor action with clear feedback
 - 2 = moderate coordination or alignment
 - 3 = fine positioning or hidden engagement
 - 4 = multiple precise coordinated actions or repeated correction
- Record whether prompting was needed
- Add step scores for each concept

Percentage of tested wrist circumference range accommodated

- Measure participant's wrist circumference at the intended wear location and ensure it lies within desired range
- Record measurement
- Apply wristband using the normal method
- Check if it closes correctly, stays secure, and has enough clearance

- Record pass only if all conditions are met
- Repeat for every participant and concept
- Divide passes by total users tested to get accommodation percentage
- Record the smallest and largest wrist sizes that passed

Wrist clearance

- Apply wristband normally
- Check two-finger gap
- Do a short movement or shake check
- Record whether wristband stays in place
- Record pass or fail
- Repeat for all concepts

Force before rupture

- Apply pull in chosen loading direction
- Add load in equal increments
- Hold each load increment for the same amount of time
- Keep increasing load until band ruptures
- Record highest load sustained
- Compare concepts using minimum failure load and average failure mode

Verification and Testing Procedures

Wristband Attachment Mechanism

1. Application time per wristband

Method

Timed usability verification using a stopwatch or video timestamp, structured around ISO 9241-11 and ISO 20282-1. ISO 9241-11 defines usability in terms of effectiveness, efficiency, and satisfaction in a specified context of use, so this method measures efficiency directly while holding context, task, and success condition constant. ISO 20282-1 is relevant because it addresses ease of operation of everyday products and requires consideration of user characteristics and context of use. For that reason, the test uses a defined start state, a defined end state, a fixed instruction, and repeated trials by representative novice users rather than informal timing by the design team.

Applicable Requirements

The requirement being verified is that application time per wristband should be as short as possible while still ending in a correct application. ISO 9241-11 drives the decision that a fast but incorrect closure cannot count as a successful result, because efficiency must be interpreted together with successful task completion in context. ISO 20282-1 drives the decision to test novice users, since ease of operation for everyday products should account for intended user characteristics rather than expert-only performance. This is an ISO-informed usability verification, not a formal ISO certification test.

Purpose/Scope

This verification activity establishes whether each concept allows users to complete the wristband application task efficiently in the intended context of use. It collects elapsed time from first touch to correct final closure, together with whether the application was completed successfully without intervention. Those observations support the conclusion because ISO 9241-11 treats usability as an outcome of use and frames efficiency in relation to successful task completion in context, while ISO 20282-1 links ease of operation to context of use and user characteristics. The rationale is therefore grounded in those usability standards, which justify controlling the task, user group, and success condition rather than relying on anecdotal observations. The conclusion is necessarily limited because the sample is small and the environment is simplified. The procedure does not prove performance under all real deployment conditions, nor does it capture long-term learning or fatigue effects.

Items Under Test

Each candidate wristband mechanism in a prototype state suitable for repeated application trials, along with the intended instruction or visual cue that would accompany normal use.

Precautions

Participants shall stop immediately if the mechanism pinches, causes discomfort, or requires excessive force. Timing shall not continue through unsafe or obviously invalid use. Remove rings, watches, and bracelets before testing.

Special Conditions/Limitations

This procedure is a summative proxy for usability efficiency, not a complete usability study. Results are valid only for the tested user group, instructions, and conditions. The method is strongest for comparing concepts under the same controlled conditions, not for claiming universal application times.

Equipment/Facilities

Stopwatch or phone timer, phone camera for audit timing, standardized wrist surrogate or live participant wrist, data sheet, and a table workspace. The simple equipment is acceptable because the standards drive the need for a controlled and repeatable context, not expensive instrumentation. If a stopwatch is used, compare it against a phone clock over a 60 s interval and confirm agreement within ± 0.2 s before testing.

Data Recording

For each trial record concept ID, participant ID, trial number, wrist circumference category, start time, end time, total time, success or failure, need for intervention, and visible cause of delay. These fields were chosen because ISO 9241-11 requires usability outcomes to be interpreted in relation to user, task, and context, not as isolated raw times.

Acceptance Criteria

A trial counts only if the wristband reaches the defined correct end state. Concepts are compared primarily by median successful application time, not best-case time, because the standards treat usability as an outcome for intended users rather than isolated ideal performance. A concept is acceptable if it achieves a median successful application time of 5 s or less and a first-attempt success rate of at least 90%. The time threshold comes from the project criterion, while the success-rate requirement was added because ISO 9241-11 does not allow efficiency to be judged separately from effective task completion.

Procedures

1. Define the exact start condition and correct end condition before testing.
2. Provide each participant the same one-sentence instruction.
3. Give one practice trial only, so the scored trial reflects walk-up usability more closely in the sense of ISO 20282-1.
4. Start the timer at first touch of the mechanism.
5. Stop the timer only when the closure is in the correct final state.
6. Mark the trial invalid if the tester intervenes physically.
7. Repeat three scored trials per participant per concept, randomizing concept order.
8. Use video review to confirm disputed times.
9. Calculate median successful time and success rate for each concept.

Troubleshooting

If timing varies widely, check whether the end condition is being judged consistently, since ISO 9241-11 requires the task outcome to be defined before efficiency can be interpreted. If one participant misunderstands the task, restate the standard instruction and repeat only that trial.

Post-Test Activities

Report median successful application time, success rate, and the mechanism feature causing the largest delay. Use the results to simplify the closure sequence, reduce alignment demands, or improve guidance features.

2. Complexity of actions per step

Method

Structured task analysis and action-complexity scoring informed by ISO 9241-110 and ISO 20282-1. ISO 9241-110 describes general interaction principles for user-system interaction, so the rubric is built around observable characteristics that make interaction harder, such as hidden state changes, need for precision, multi-action coordination, and poor self-descriptiveness. ISO 20282-1 supports evaluating ease of operation for everyday products with respect to user characteristics and context of use. Because there is no exact ISO scoring method for a mechanical wristband action sequence, this is explicitly ISO-inspired rather than a direct compliance test.

Applicable Requirements

The requirement is that the mechanism should involve simple actions per step. ISO 9241-110 drives the decision to evaluate step complexity in terms of interaction principles rather than personal opinion. For example, steps are penalized when they are not self-descriptive, require unnecessary coordination, or depend on fine alignment without clear feedback. ISO 20282-1 drives the decision to use novice participants and to document the context of use.

Purpose/Scope

This verification activity establishes whether one concept requires a simpler and more transparent user action sequence than another. It records the number of task steps, the number of elemental actions per step, and a structured complexity score based on whether the step requires fine alignment, simultaneous actions, high precision, hidden engagement, or repeated correction. Those observations support the conclusion because ISO 9241-110 frames good interaction in terms of principles such as self-descriptiveness and controllability, and ISO 20282-1 ties ease of operation to the interaction demands imposed on intended users. The rationale is therefore grounded in the standards' focus on interaction quality and ease of operation. The conclusion remains limited because scoring still involves judgment, even with a rubric. The procedure does not directly measure cognitive load or long-term learning.

Items Under Test

Each wristband prototype, plus any instruction card or visible cue included in intended use.

Precautions

Use two observers and a common rubric to reduce scorer bias. Do not coach the user during the scored trial. Stop testing if force or posture becomes uncomfortable.

Special Conditions/Limitations

This is a proxy test because the standard provides design principles, not a mandatory numerical scoring scheme for this product type. The result is valid mainly for comparing concepts using the same rubric and user group.

Equipment/Facilities

Video camera, clipboards, printed step rubric, prototypes, and a standard workspace. No advanced equipment is required because the standards driving this method concern interaction quality and user operation, not force or motion capture. Before testing, both observers should jointly score two pilot trials and reconcile differences so their scores align within 1 point.

Data Recording

Record the ordered task steps, elemental actions per step, whether the step was self-evident or required prompting, whether simultaneous coordination was needed, whether correction occurred, and each observer's score. These data items were selected directly from the interaction concerns emphasized by ISO 9241-110 and the ease-of-operation framing in ISO 20282-1.

Acceptance Criteria

A step is scored 1 if it is a single obvious gross-motor action with clear feedback, 2 if it requires moderate coordination or alignment, 3 if it requires fine positioning or hidden engagement, and 4 if it involves multiple coordinated precise actions or repeated correction. Lower total score is better. A concept is acceptable if no step averages above 3 and the total average score is lower than competing concepts while maintaining successful application. These acceptance rules come from ISO 9241-110's interaction principles, especially self-descriptiveness and controllability, which justify penalizing steps that are obscure, correction-prone, or hard to control.

Procedures

1. Define the standard task start and end states.
2. Observe one full application trial per participant after one familiarization run.
3. Break the interaction into discrete user-visible steps.
4. Score each step using the rubric immediately after observation.
5. Review video for disagreements.
6. Sum step scores and count elemental actions.
7. Compare concepts by mean total score and highest-scoring step.

Troubleshooting

If observers disagree frequently, redefine the step boundaries and anchor examples more clearly, since the standards support principled evaluation but do not provide a ready-made rubric for this exact mechanism.

Post-Test Activities

Identify the step with the highest interaction burden and redesign it to improve self-descriptiveness, reduce precision demand, or eliminate simultaneous actions.

3. Percentage of tested wrist circumference range accommodated

Method

Anthropometric fit verification using wrist circumference measurements and fit trials, driven by ISO 7250-1 and ISO 15535. ISO 7250-1 provides definitions and landmarks for body measurements used in technological design. ISO 15535 requires that anthropometric data be reported together with population characteristics, sampling method, and measurement items. That is why this procedure specifies a measurement location, requires the intended user population to be stated, and records pass or fail by wrist size rather than reporting only a single overall percentage.

Applicable Requirements

The requirement is that the design accommodate as high a percentage of intended users as possible. ISO 7250-1 drives the decision to base accommodation on explicit body measurement definitions rather than informal size labels. ISO 15535 drives the decision to report user population, sampling basis, and

measured statistics, since anthropometric conclusions are not meaningful without that context. This is an ISO-inspired fit test, not an anthropometric database study.

Purpose/Scope

This verification activity establishes the fraction of the intended wrist-size range that can be fitted securely and comfortably by each concept. It uses wrist circumference measured with a defined method, together with whether the mechanism closes, remains secure, and still meets the clearance criterion on each participant or surrogate. Those observations support the conclusion because ISO 7250-1 frames body measurements as the basis for geometric design decisions, and ISO 15535 requires those measurements and the associated population description to be reported in a way that supports valid comparison. The rationale is therefore grounded in anthropometric design practice rather than informal size judgments. The conclusion is limited because the sample is small and likely opportunistic rather than population-representative. The procedure does not prove accommodation of all real users unless the tested range genuinely reflects the target user population.

Items Under Test

Each prototype and its adjustable range, tested on human wrists or cylindrical surrogates spanning the intended wrist circumference range.

Precautions

Measure only over bare skin and do not overtighten. If a live participant reports discomfort, stop the trial.

Special Conditions/Limitations

A convenience sample may reveal comparative fit trends, but anthropometric conclusions depend on how the population and sampling are defined and reported. If surrogates are used, the test becomes a geometric proxy rather than a human accommodation study.

Equipment/Facilities

Flexible tape measure, ruler, wrist surrogates of known circumference, prototypes, and data sheets. These are sufficient because the standards require consistent body-dimension definition and reporting, not expensive instrumentation. Compare the flexible tape against a rigid ruler over 100 mm and confirm deviation within ± 1 mm before use.

Data Recording

Record participant or surrogate ID, intended user group, wrist circumference, concept ID, closure success, retention success, clearance success, comfort pass or fail, and overall accommodation pass or fail. This structure follows ISO 15535's emphasis on reporting the relevant population, measurements, and resulting statistics.

Acceptance Criteria

A test subject counts as accommodated only if the mechanism closes correctly, stays secured, and passes the wrist-clearance requirement. Mere closure without functional fit does not count. The accommodation percentage is reported as the number of fully passing subjects divided by the number tested in the defined target range. If the team has a target percentile range, the concept should cover that range. If not, concepts should be compared by the widest fully passing circumference range and highest full-pass percentage. This use of a measured range rather than a single anecdotal pass is directly driven by the anthropometric standards.

Procedures

1. Define the intended user group and wrist-size range before testing.
2. Measure circumference at the intended wear location using the same method for all subjects.
3. Order subjects from smallest to largest wrist size.
4. Fit each concept using the normal application procedure.
5. Check closure, retention, and clearance.
6. Record full pass only if all required fit conditions are met.
7. Plot pass or fail against wrist circumference.

Troubleshooting

If repeated wrist measurements differ, repeat until two readings agree closely and use their average. If failures cluster only at one end of the size range, document whether the cause is length, adjustment increment, or closure geometry.

Post-Test Activities

Report the passing circumference range and the pass percentage for the tested population. Use the results to revise adjustable range or sizing increments.

4. Wrist clearance

Method

Dimensional inspection using go or no-go thickness gauges after application, shaped by ISO 7250-1 and ISO 14738. ISO 7250-1 provides the anthropometric measurement basis for relating product dimensions to body dimensions. ISO 14738 establishes principles for deriving design dimensions from anthropometric measurements and applying them to physical design. There is no exact ISO procedure for checking the gap under a wristband with paper gauges, so this is an ISO-inspired dimensional proxy. The standards drive the choice to use a defined geometric band of acceptable clearance rather than a vague statement such as 'not too tight.'

Applicable Requirements

The requirement is adequate clearance without looseness. ISO 14738 drives the decision to convert this qualitative need into dimensional pass and fail limits, since ergonomic design requires derived dimensions, not only verbal impressions. ISO 7250-1 drives the decision to check clearance relative to actual wrist size at the intended wear location.

Purpose/Scope

This verification activity establishes whether each concept maintains a controlled gap between band and wrist that is large enough to avoid over-tightness but small enough to avoid excessive looseness. It records minimum and maximum gauge insertion results at defined wrist locations, together with a short migration check after movement. Those observations support the conclusion because ISO 14738 is concerned with deriving acceptable design dimensions from anthropometric considerations, while ISO 7250-1 provides the body-measurement basis for that derivation. The rationale is therefore grounded in dimensional ergonomic design rather than subjective feel alone. The conclusion is limited because soft tissue compliance and local geometry affect the apparent gap. The procedure does not directly measure pressure distribution or long-duration comfort.

Items Under Test

Applied wristband prototypes on live wrists or compliant surrogates covering the intended size range.

Precautions

Insert gauges gently and do not force them under the band. Stop if the participant reports discomfort.

Special Conditions/Limitations

Because no exact ISO wristband-clearance method was verified, this is a dimensional proxy built from anthropometric-design principles. It is suitable for concept comparison and requirement checking only if the target clearance band has been defined beforehand.

Equipment/Facilities

Stacked paper or cardboard gauges of measured thickness, ruler or calipers, tape measure, and data sheets. These tools are adequate because the standards driving the method concern dimensional derivation and anthropometric fit, not high-resolution sensing. Measure each gauge stack three times and label the average thickness to the nearest 0.5 mm.

Data Recording

Record wrist circumference, gauge thicknesses, whether the minimum gauge enters at each location, whether the maximum gauge enters, and whether the band migrates during the shake check. These fields were chosen so the final result is based on explicit dimensional evidence rather than general comfort comments alone.

Acceptance Criteria

The acceptable clearance band must be specified before testing. Acceptance is based on a derived dimensional range, not a subjective judgment. A trial passes if the minimum gauge enters at all specified locations, the maximum gauge does not enter, and the band does not migrate excessively during light movement. The exact gauge sizes should be justified from the team's ergonomic requirement, but once set they must remain fixed across concepts. This structure comes directly from ISO 14738's dimensional-design logic.

Procedures

1. Define the target clearance band from the team requirement.
2. Build and label minimum and maximum gauges.
3. Apply the wristband normally.
4. Check two finger gap.
6. Perform a short movement or shake check.
7. Record pass or fail and the location of any failure.

Troubleshooting

If gauge insertion itself disturbs the fit, reduce the leading-edge stiffness of the gauge or standardize the insertion angle. If different testers disagree, demonstrate a common insertion force and orientation before resuming.

Post-Test Activities

Use failed locations to identify where contact geometry or adjustment increment needs revision.

5. Force before rupture

Method

Product-level tensile loading test inspired by ISO 527-1. ISO 527-1 specifies general principles for determining tensile properties under defined conditions. Although the standard is written for plastics and plastic composites with defined specimens, it still drives the core choices here: load shall be applied in a defined direction, the loading setup shall be consistent across specimens, the force at failure shall be recorded, and failure mode shall be documented. Because the student project uses the actual wristband mechanism and a low-cost hanging-load rig rather than a universal test machine, this is explicitly ISO-inspired and not a compliance tensile test.

Applicable Requirements

The requirement is that force before rupture should be as high as possible. ISO 527-1 drives the decision to define a consistent loading direction, a controlled incremental loading method, and failure as the endpoint rather than stopping at first visible deformation. It also drives the requirement to test multiple specimens, since tensile conclusions should not be based on a single sample.

Purpose/Scope

This verification activity establishes the maximum tensile load each concept can sustain before rupture or catastrophic loss of function. It records the applied load at failure, the loading direction, visible deformation before failure, and the failure mode and location. Those observations support the conclusion because ISO 527-1 is built around the principle that tensile performance must be determined under defined loading conditions with recorded failure outcomes. The rationale is therefore grounded in tensile-testing logic, even though the apparatus is simplified. The conclusion is limited because the setup is not a laboratory universal testing machine and the specimen is the finished mechanism rather than a standard coupon. The procedure does not yield material properties such as tensile strength in MPa, and it does not predict fatigue or environmental aging.

Items Under Test

At least five specimens of each prototype concept, all tested in the same orientation and fixture.

Precautions

Use eye protection and shield the falling load path. Keep hands and feet clear and do not exceed the fixture's safe load.

Special Conditions/Limitations

This is a product-level rupture test, not a standardized material test. The force result is suitable for concept comparison only if the fixture geometry, loading rate, and orientation are kept constant.

Equipment/Facilities

Rigid support frame, consistent loading fixture, bucket or weight hanger, measured masses or water bottles, optional spring scale, ruler, and camera. This equipment preserves the standard's key principle of defined loading conditions while remaining realistic for a first-year project. If a spring scale is used, verify it against at least one known mass before testing. If water is used, treat 1 L as approximately 1 kg and label each bottle volume clearly.

Data Recording

Record specimen ID, concept ID, fixture orientation, mass increments, corresponding load, hold duration, visible deformation, failure load, failure mode, and failure location. These data fields are directly justified by ISO 527-1's general principles for tensile determination under defined conditions.

Acceptance Criteria

Each specimen shall survive a defined service-load proxy before being considered viable, and concepts shall then be ranked by minimum failure load rather than only mean failure load. That choice is driven by the standard's focus on defined conditions and repeatability, and by engineering safety logic: a concept with one weak specimen is less robust even if its average is high. If the team has a required service load, a safety factor should be applied to that value and every specimen should exceed it. Otherwise concepts should be compared by minimum and average rupture load and by failure mode.

Procedures

1. Define the pull direction that best represents opening or accidental tensile loading in use.
2. Mount every specimen in the same fixture and orientation.
3. Add load in equal increments.
4. Hold each increment for a fixed time.
5. Continue until rupture or complete functional failure.
6. Record the highest sustained load and the failure mode.
7. Repeat for at least five specimens per concept.

Troubleshooting

If failures occur in the fixture rather than the prototype, redesign the fixture and restart the series, because the defined load path must act on the specimen rather than on an uncontrolled weak point in the setup.

Post-Test Activities

Photograph failures and categorize them so redesign is based on the actual rupture location and mechanism.

6. Force applied to wrist

Method

Low-cost proxy evaluation of wrist loading using measured tightening force, contact width, and immediate comfort or marking check, informed by ISO 9241-11, ISO 20282-1, and anthropometric design logic from ISO 14738. No verified ISO standard was found that gives a simple student-level method for directly measuring local wrist pressure from this kind of wearable mechanism. For that reason, the standards are used to shape the proxy rather than to claim direct compliance. Specifically, ISO 9241-11 and ISO 20282-1 justify measuring the interaction outcome on intended users in context, while ISO 14738 supports relating body contact to physical dimensions and force paths.

Applicable Requirements

The requirement is that force applied to the wrist should be as low as possible while the band still closes securely and maintains required clearance. The standards drive the decision that wrist loading cannot be judged from force alone. Because usability is an outcome in context, the proxy combines tightening force, contact width, retention, clearance, and immediate discomfort. This prevents selecting a concept that is comfortable only because it is too loose to function.

Purpose/Scope

This verification activity establishes which concept achieves the required fit and retention with the lowest plausible wrist-loading proxy. It records peak tightening force during application, width of the

contact region, the resulting force-per-width proxy, and immediate user response such as discomfort or visible temporary marking. Those observations support the conclusion because local loading on the wrist is influenced by the force needed to tension the mechanism and the area over which that load is distributed, while ISO 9241-11 and ISO 20282-1 require that such measures be interpreted in the real use context. ISO 14738 further supports using geometric contact information to relate body dimensions to design decisions. The rationale is therefore grounded in usability and ergonomic design principles, even though the exact tissue pressure is not directly measured. The conclusion is limited because this is not a direct pressure measurement. The procedure does not prove medical safety, long-term circulation effects, or detailed pressure distribution.

Items Under Test

Complete wristband prototypes tested on the intended wrist location with the same closure target used in the clearance test.

Precautions

Stop immediately if there is pain, numbness, tingling, or skin pinching. Keep wear duration short.

Special Conditions/Limitations

This is explicitly a proxy method. The standards shape the need for contextual, user-centered evaluation, but no exact ISO method was verified for direct wrist-pressure testing with this equipment. The results are most defensible for relative comparison between concepts tested under the same closure criterion.

Equipment/Facilities

Spring scale, ruler or calipers, tape measure, phone camera, prototypes, and data sheet. These tools are enough to produce a repeatable proxy because the standards require traceable context and defined physical conditions, not advanced pressure-mapping hardware. Verify the spring scale against at least two known hanging masses and confirm readings within $\pm 5\%$ over the expected range.

Data Recording

Record concept ID, wrist size, closure position, peak tightening force, contact width, force-per-width proxy, clearance pass or fail, retention pass or fail, discomfort score, and visible marking duration. These data were selected because the standards require the measured outcome to be tied to use context and user response, not only to a mechanical input value.

Acceptance Criteria

A concept shall only be judged acceptable on wrist-force grounds if it first passes clearance and retention. Among concepts that do so, the one with the lower median force-per-width proxy and lower discomfort is preferred. This acceptance logic is driven directly by ISO 9241-11 and ISO 20282-1, which do not support judging comfort or efficiency apart from effective use in context. The no-excessive-marking condition is included because it provides a visible supporting indicator that the required fit may be imposing unnecessary local load.

Procedures

1. Measure contact width at the main load-bearing region.
2. Attach the spring scale inline with the tightening direction.
3. Tighten only until the band reaches the defined acceptable closure state that also satisfies clearance.
4. Record peak tightening force.

5. Complete closure without further tightening.
6. Record retention result, clearance result, discomfort score, and any temporary marking.
7. Calculate force-per-width for concept comparison.
8. Repeat across several wrist sizes.

Troubleshooting

If the scale changes the application path, insert a consistent small loop at the same point on every prototype. If force readings fluctuate, slow the pull rate and align the scale with the strap direction.

Post-Test Activities

Use the results to redesign contact width, closure travel, or locking geometry so acceptable retention can be achieved with lower tightening input.